Retail Analytics Metrics Glossary

This document defines the business metrics used in customer and segment-level retail analytics. It is the canonical reference for *what a metric means and how it should be calculated*. It does not specify physical table or column names — match the logic below against the live database schema (see the schema tool) before writing SQL, since column names and exact status values can differ from what's described here.

Customer Lifetime Value - Pre-Discount (LTV Gross)

Definition: The total gross revenue a customer has generated across their entire order history, valued at original list price before any discounts, coupons, or promotional reductions are applied.

Calculation logic: Sum the pre-discount line-item or order-level amount across all of a customer's orders.

Inclusion / exclusion rules:

- Cancelled orders (orders that never shipped) are excluded — they generated no actual transaction and should not count toward gross value.
- Returned orders ARE included in pre-discount LTV. This metric reflects original purchase intent and gross demand generated by the customer, regardless of what happened to the order afterward. Whether a customer later returned an item is a separate signal, not a reason to erase the original sale from this metric.
- Shipping fees and taxes are excluded; this metric reflects product revenue only.

Notes: This is a "demand-side" metric — it answers "how much did this customer attempt to buy," not "how much revenue did the business actually keep."

Customer Lifetime Value - Post-Discount (LTV Net)

Definition: The total net revenue actually realized from a customer, after discounts and after netting out refunds from returned orders.

Calculation logic: Sum of (order amount - discount amount) across all orders, MINUS the refunded amount for any orders that were returned.

Inclusion / exclusion rules:

- Cancelled orders are excluded, same as in the pre-discount metric.
- Returned orders are included but their value is netted to the refunded amount (typically zero for a full return, or the residual amount for a partial return/ partial refund). This is the key difference from LTV Pre-Discount: a returned order should reduce realized value here, even though it still counts toward gross demand above.
- If a return was processed as store credit rather than a cash refund, treat

it as netted out the same way — store credit is not realized product revenue at the point of return.

Notes: This is the "revenue-side" metric — it answers "how much value did this customer actually leave on the books." LTV Gross minus LTV Net for a given customer is, by construction, approximately equal to their total discounts given plus their total refunded amount.

Return Rate

Definition: The proportion of a customer's (or segment's) orders that were returned, out of their total completed orders.

Calculation logic: Return Rate = (count of orders with a return) / (count of total orders), evaluated at the order level, not the line-item level. A multi-item order with one returned item still counts as one returned order under this definition unless a line-item-level return rate is explicitly requested instead.

Inclusion / exclusion rules:

- • An order counts as "returned" only if the customer received a refund or credit after fulfillment — i.e., the item shipped and was sent back. This is distinct from a cancellation, which happens before fulfillment and reflects a different customer behavior (changed their mind before the order shipped, not after receiving it). Cancelled orders must never be counted as returns, and should typically also be excluded from the denominator, since they were never a completed transaction in the first place.
- • The denominator should be total *completed* orders (i.e., orders that shipped), not all orders ever placed, to avoid diluting the rate with orders that were cancelled pre-shipment.
- • If an order has multiple status changes over its lifecycle (e.g., shipped → returned), use the most recent/current status to determine whether it counts as returned, not an intermediate one.

Notes: This is the metric most likely to be silently wrong if you trust a guessed status string instead of checking the actual distinct values present in the data — always verify the exact status label used for a return before filtering on it.

Recent Purchase Activity (30 / 60 / 90-Day Purchase Count)

Definition: The number of completed orders a customer has placed within a trailing window of 30, 60, or 90 days from the reference date (normally the current date, or a specified analysis date for historical snapshots).

Calculation logic: For each window, count orders where the order date falls within [reference date - N days, reference date], for N in {30, 60, 90}. Each window is independent and inclusive of the same orders as the shorter windows it contains (i.e., the 90-day count always includes whatever happened in the 30-day count).

Inclusion / exclusion rules:

- • Only orders that actually shipped/completed count toward this metric. Cancelled orders do not represent purchase engagement and must be excluded.
- • Returned orders ARE still counted here. A return is a post-purchase event; the act of placing and completing the order is what this metric measures, not whether the customer ultimately kept the item. (Contrast this with LTV Net, where a return reduces realized value — recency of engagement and realized revenue are different questions and should not be conflated.)
- • Use order date (the date the order was placed), not ship date or delivery date, as the anchor for window membership, unless a specific analysis explicitly calls for fulfillment-date windows instead.

Notes: This metric is typically used as a recency/engagement signal (e.g., feeding into churn risk or reactivation segmentation), not as a revenue metric.

Category Affinity

Definition: The product category that a customer purchases from most, and the strength of their concentration in that category relative to their overall purchase activity.

Calculation logic: For each customer, group their order history by product category and sum either order count or revenue per category (revenue is the default basis unless count is explicitly requested). The customer's "affinity category" is the category with the highest total. The "affinity score" is that category's share of the customer's total activity: affinity score = (value in top category) / (value across all categories).

Inclusion / exclusion rules:

- • Returned items are excluded from the category affinity calculation — a returned purchase does not reflect a genuine preference for that category, since the customer ultimately rejected the item. This differs from the LTV Gross and Recent Purchase Activity metrics above, where returns are still counted; affinity is specifically about retained preference, not raw transaction volume.
- • Cancelled orders are excluded for the same reason as elsewhere in this document.
- • A customer with only one order has, by definition, a 100% affinity score toward whatever category that order falls into — this is expected and not a data quality issue, but should be flagged or filtered out separately if the analysis requires a minimum order count for a meaningful affinity signal (a common threshold is 3+ completed orders).

Notes: Category affinity is typically used for segment-level merchandising and personalization analysis, not as a financial metric.